Miscellaneous problems

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1 Multiple choice questions

There can be any number of answers to each question, unless otherwise specified.

1. Suppose $\mathcal{M}_\Gamma$ is a canonical model for Γ (in whichever logic). Which of the following must be true?
   (a) $\mathcal{M}_\Gamma \models \Gamma$.
   (b) If $\Gamma \vdash \varphi$, $\mathcal{M}_\Gamma \models \varphi$.
   (c) If $\mathcal{M}_\Gamma \models \varphi$, then $\Gamma \vdash \varphi$.
   (d) There is no smaller model of Γ than $\mathcal{M}_\Gamma$.
   (e) If $\Gamma = \emptyset$, then $\mathcal{M}_\Gamma$ does not satisfy any sentences.

2. Suppose $p, q, r \in P$ and $\{\mathsf{All}\ p\ \mathsf{are}\ q, \mathsf{All}\ q\ \mathsf{are}\ r\} \vdash \mathsf{All}\ q\ \mathsf{are}\ p$. Which of the following statements must be correct? (Note that for any two members of the same set, we may ask whether they're identical or not, so the following choices are meaningful.)
   (a) $p = q$
   (b) $p = r$
   (c) $q = r$
   (d) $p \neq q$
   (e) $p \neq r$
   (f) $q \neq r$

3. Suppose that $\Gamma \vdash \varphi$ and $\Gamma \cup \{\varphi\} \vdash \psi$. Which of the following must be true? (We can interpret $\vdash$ as derivability in any proof system—it doesn't actually matter which one we choose! Pick your favorite one.)
   (a) $\Gamma \vdash \psi$
   (b) $\Gamma \cup \{\psi\} \vdash \varphi$
   (c) $\Gamma \cup \{\varphi\} \vdash \varphi$
   (d) $\{\varphi, \psi\} \vdash \psi$

4. Suppose $\Gamma \subseteq \Delta$. Which of the following are true? (Again, we can work in any logic—pick your favorite.)
   (a) If $\Gamma \models \varphi$ then $\Delta \models \varphi$, for any sentence φ.
   (b) If $\Delta \models \varphi$ then $\Gamma \models \varphi$, for any sentence φ.

5. Now work in propositional logic specifically. If $\{\chi, \varphi\} \models \psi$ and $\{\chi, \psi\} \models \varphi$, which of the following must be true?
   (a) $\varphi \vee \psi$ is satisfiable
   (b) $(\varphi \leftrightarrow \psi) \to \chi$ is satisfiable

(c) $\chi \to (\varphi \leftrightarrow \psi)$ is a tautology
(d) $\chi \wedge \neg\psi \wedge \varphi$ is a contradiction

6. Suppose we have a logic with model-theoretic entailment ($\models$) and proof-theoretic derivability ($\vdash$), the latter being defined as the existence of some proof tree each of whose nodes satisfies some rule. Suppose this system is sound and complete, viz., $\Gamma \models \varphi \iff \Gamma \vdash \varphi$. Now suppose I *change* the definition of $\vdash$ by *adding* some extra proof rules, to get $\vdash'$. Which of the following are true? Pick exactly one answer.

   (a) The logical system is still complete (if $\Gamma \models \varphi$ then $\Gamma \vdash' \varphi$), but is possibly unsound
   (b) The logical system is still sound (if $\Gamma \vdash' \varphi$ then $\Gamma \models \varphi$), but is possibly incomplete
   (c) The logical system must be sound and complete
   (d) The logical system is possibly unsound and incomplete

7. Which of the following first-order formulas says exactly that "there are at least two elements in the universe?"

   (a) $(\exists x, y)(x = y)$
   (b) $(\forall x, y)(x \neq y)$
   (c) $(\forall x)(\exists y)(x \neq y)$
   (d) $(\exists x, y)(x \neq y)$

8. Consider the ring of integers $(\mathbb{Z}, 0, 1, +, \cdot)$ and the ring of real numbers $(\mathbb{R}, 0, 1, +, \cdot)$ as first-order structures. Which of the following sentences are satisfied by exactly one of these structures, and falsified by the other? (In other words, on which sentences do these two structures disagree?)

   (a) $(\forall x)(\exists y)(x + y = 0)$
   (b) $(\exists y)(\forall x)(x + y = 0)$
   (c) $(\exists x)(x \cdot x = 1)$
   (d) $(\exists x)((x \cdot x) + 1 = 0)$

9. The boolean operator $\oplus$ is *exclusive or*, so $p \oplus q$ is true exactly when *one of* p and q are true and the other is false. What would be a correct definition of $p \oplus q$ in terms of what we already know?

   (a) $\neg(p \leftrightarrow q)$
   (b) $(p \wedge \neg q) \vee (\neg p \wedge q)$
   (c) $(p \to \neg q) \wedge (q \to \neg p)$
   (d) $(p \vee q) \wedge (\neg p \vee \neg q)$

10. Work over models in the signature $\{p, q, r\}$. How many models of $(p \leftrightarrow q) \vee (q \leftrightarrow r) \vee (r \leftrightarrow p)$ are there?

    (a) 5
    (b) 6
    (c) 7
    (d) 8

11. Work in the logic $\mathcal{A}$. Suppose my set of nouns is $P = \{p_0, p_1, p_2, \dots\}$s. Suppose Γ is the collection of all sentences of the form All p_{i+1} are p_i—so All p_1 are p_0, All p_2 are p_1, etc. Which of the following statements are correct?

    (a) $\Gamma \cup \{\text{All } p_{100} \text{ are } p_{1000}\} \models \text{All } p_{200} \text{ are } p_{2000}$
    (b) $\Gamma \cup \{\text{All } p_{200} \text{ are } p_{2000}\} \models \text{All } p_{100} \text{ are } p_{1000}$

(c) $\Gamma \cup \{\text{All } p_{100} \text{ are } p_{2000}\} \models \text{All } p_{200} \text{ are } p_{1000}$

(d) $\Gamma \cup \{\text{All } p_{200} \text{ are } p_{1000}\} \models \text{All } p_{100} \text{ are } p_{2000}$

12. Which of the following statements correctly expresses the negation of "at least two-thirds of computer science students have had prior programming experience?"

    (a) At least two-thirds of computer science students have not had prior programming experience.
    (b) At least one third of computer science students have not had prior programming experience.
    (c) At most two-thirds of computer science students have had prior programming experience.
    (d) At most two-thirds of computer science students have not had prior programming experience.

2 Short-answer questions

These are generally of the length and scope that I could reasonably ask on the final exam—maybe some are stretching it a little bit, but not too much.

1. Is there a transformation $M \mapsto M^\star$ on models of propositional logic such that $M \models \neg\varphi \iff M^\star \models \varphi$? Either define one or prove no such thing exists.

2. Let's say I have a transformation on sentences that I am arguing *preserves provability*. By "preserves provability" I mean that if I have an instance of $\Gamma \vdash \varphi$ and I transform each sentence in Γ and also φ, then the latter is still provable from the former. An example of this are the *dual* sentences we talked about: remember that we proved the statement

   $$\text{For all } \Gamma, \varphi, \text{ if } \Gamma \vdash \varphi \text{ then } \Gamma^\dagger \vdash \varphi^\dagger.$$

   We use *induction on proof trees* to prove statements like this. But induction on proof trees is used to prove statements of the form "for all trees ..." and that doesn't seem to match the form of the statement above! What gives? Give a clear and concise explanation of what's going on.

3. Let's say formulas φ and ψ are *co-satisfiable* in case there is a model that satisfies *both* φ and ψ. Is co-satisfiability a transitive relation? (Work in propositional logic.)

4. Build a propositional model over the signature $\{p, q\}$ with exactly three satisfying assignments.

5. Do exercise 2.6 in the book (which concerns canonical models in $\mathcal{A}(\mathcal{RC})$).

6. There is a translation of $\mathcal{A}(\mathcal{RC})$-terms into first order terms as follows. For example, take the $\mathcal{A}(\mathcal{RC})$ signature with nouns hawks, turtles and verbs sees, spies. We get a first-order signature with unary relation symbols hawks, turtles and a binary relation symbols sees, spies. In this way each $\mathcal{A}(\mathcal{RC})$ model *is* a first-order model.

   Now we want to define a map from terms to terms so that the interpretation of the term of the $\mathcal{A}(\mathcal{RC})$-term on the left is identical to the interpretation of the first-order term on the right, like the following examples show. (Notice that I've made the free variable x in each case.)

hawks	$\text{hawks}(x)$
sees all hawks	$(\forall y)(\text{hawks}(y) \to \text{sees}(x, y))$
spies all sees all turtles	$(\forall y)(((\forall z)(\text{hawks}(z) \to \text{sees}(y, z))) \to \text{spies}(x, y))$

   Can you generalize this pattern by defining an interpretation-preserving map from $\mathcal{A}(\mathcal{RC})$-terms to first-order terms in an *arbitrary* signature? (And, of course, prove that this map *does* in fact preserve the interpretation?)

7. Consider the $\mathcal{A}(\mathcal{RC})$ signature with a single noun z and a single verb ℓ. This means that all terms have the form $\ell^{(n)}\, z$ for some natural number n: z, ℓ all z, ℓ all (ℓ all z), etc. Consider a model in this signature whose domain is $\mathbb{Z}$, the set of integers, and where z is interpreted by $\emptyset$, the empty set, and ℓ is interpreted by the less-than relation, so $[\![\ell]\!] = \{(x, y) : x < y\}$.

   For each n, calculate the denotation $[\![\ell^{(n)}\, z]\!]$ of the term $\ell^{(n)}\, z$.

8. Find and justify tableau rules for $\to$ and $\leftrightarrow$.

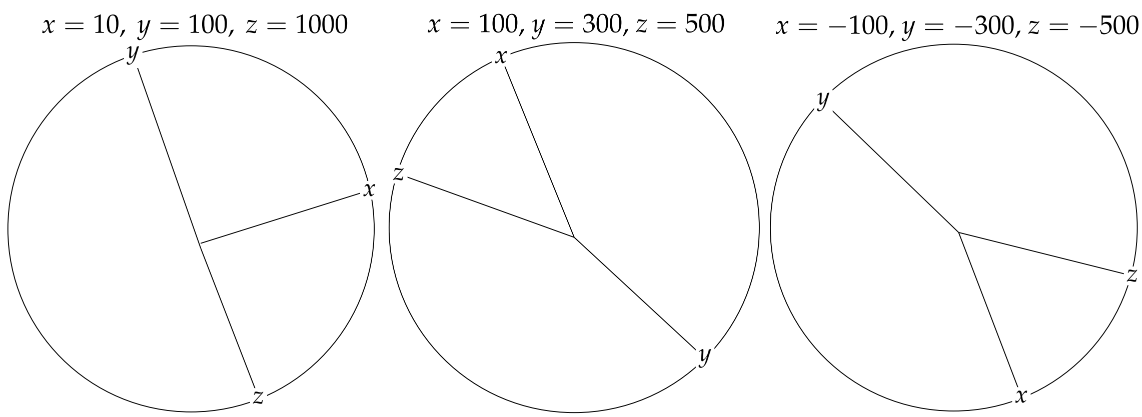
9. Having done the previous determine whether $((p \to q) \to p) \to p$ is a tautology or falsifiable via the method of tableaux. In case of the latter, find a falsifying assignment.

10. Determine whether $(\forall x, y)(L(x, y) \to L(y, x)) \models (\forall x, y)(L(x, y) \leftrightarrow L(y, x))$. If true, prove that any model satisfying the left-hand side must satisfy the right-hand side. If false, find a model that satisfies the left but not the right-hand sides.

11. Let Γ be a set of $\mathcal{A}(\mathcal{RC})$ sentences in some fixed signature. Is it possible to find a model which satisfies all sentences in Γ? Either prove that it is or find a concrete Γ for which it isn't.

12. I have a non-terminating program (like an operating system or similar) with a boolean-valued variable `test`. At each time step $n \in \mathbb{N}$, the value of `test` at time n is either T or F . The value of test can switch between true and false as often as you like, but to pass it has to *eventually* stabilize on "true." Express this condition as a first-order formula. (The only relation you really need is $<$.)

13. Suppose x, y, and z are three real-numbered variables. Interpret them as degree angles, so we get pictures like the following:



Write a first-order formula that captures exactly when the sequence (x, y, z) occurs in **clockwise order.** So this formula should be satisfied by $(100, 300, 500)$ and $(-100, -300, -500)$ and falsified by $(10, 100, 1000)$. You can use any algebraic operations you like (e.g., $+, \cdot, -$) as well as $<$. Your solution must be **simple and understandable.**

3 Build-your-own logic

The questions here are all of the form, "here's a partially or imperfectly described logical system. Complete it by specifying a notion of proof and/or notion of model and proving soundness and completeness theorems." Questions 3 and 4 in particular have a similar flavor: I take one of the logics from class and restrict the class of models somehow; how do I adjust the proof system to compensate?

These questions are generally *more involved* than would be reasonable to ask on a 4-hour exam, but I could absolutely draw inspiration for them or ask a special case, plus going through these is excellent practice.

1. One of the most famous classical syllogisms is the following:

$$\frac{\textsf{All men are mortal} \qquad \textsf{Socrates is a man}}{\textsf{Socrates is a mortal}}$$

Now this *seems* like it should be easily formalized in the logic $\mathcal{A}$ but that's not quite the case! This logic allows us to talk about nouns like $\textsf{man}$ and $\textsf{mortal}$, but $\mathcal{A}$ has no mechanism for *naming individuals* such as $\textsf{Socrates}$.

In this exercise, your task is to *extend* the logic $\mathcal{A}$ to accommodate inferences like the above. Signatures should consist of two sets, a set P of *nouns* and a set N of *names*. There should be two types of sentences: $\mathsf{All}\ p\ \mathsf{are}\ q$ where $p, q \in P$ and $n\ \mathsf{is\ a}\ p$, where $n \in N$ and $p \in P$.

(a) What is the correct definition of a (P, N)-*model*? For a model $\mathcal{M}$ and sentence φ, what is the correct definition of $\mathcal{M} \models \varphi$?

(b) Can you extend the *proof system* A with an additional proof rule that enables inferences such as the above? Can you prove that it's sound?

(c) (Challenging) Prove the resulting proof system is *complete*.

(d) (Challenging) Extend your Haskell program for $\mathcal{A}$ to this expanded logic.

2. Exercises 2.13-2.15 in the book develop a logic around sentences of the form $\mathsf{All}\ x\ \mathsf{which\ are}\ y\ \mathsf{are}\ z$. Do them.

3. Work in the logic $\mathcal{A}$. We say a model $\mathcal{M}$ in the signature P is *reflective* if, for any $p, q \in P$, whenever $[\![p]\!] \subseteq [\![q]\!]$, then $[\![q]\!] \subseteq [\![p]\!]$. Define $\Gamma \models^r \varphi$ if for every *reflective* model $\mathcal{M}$, if $\mathcal{M} \models \Gamma$ then $\mathcal{M} \models \varphi$.

   (a) Does $\models^r$ entail *more or fewer things* than $\models$? In other words, if $\Gamma \models \varphi$, must $\Gamma \models^r \varphi$? What about conversely? For each direction, either prove the implication, or find an entailment that is valid in the first setting but not the second.

   (b) Based on your experiments above, suggest a *proof system* corresponding to $\models^r$. Prove that it's sound and complete.

4. Work in the logic $\mathcal{A}(\mathcal{RC})$. We say a model $\mathcal{M}$ in the signature (P, R) is *symmetric* in case the interpretation $[\![r]\!]$ of every verb $r \in R$ is a symmetric binary relation on M, viz., for all $x, y \in M$ $(x, y) \in [\![r]\!] \iff (y, x) \in [\![r]\!]$. Define $\Gamma \models^s \varphi$ if for every *symmetric* model $\mathcal{M}$, if $\mathcal{M} \models \Gamma$ then $\mathcal{M} \models \varphi$.

   (a) Does $\models^s$ entail *more or fewer things* than $\models$? In other words, if $\Gamma \models \varphi$, must $\Gamma \models^s \varphi$? What about conversely? For each direction, either prove the implication, or find an entailment that is valid in the first setting but not the second.

   (b) Based on your experiments above, suggest a *proof system* corresponding to $\models^s$. (Challenging) Prove that it's sound and complete.